Internet Access and Economic Development:

Evidence from Global Panel Data

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1 Introduction

Digital connectivity is now a central feature of modern economic life. Firms use online platforms to reach customers, coordinate suppliers, advertise products, process payments, and manage logistics. Households use the internet to search for jobs, access information, communicate, and obtain public or private services. Governments increasingly rely on digital systems for taxation, procurement, social protection, and public communication. Because the internet affects how information moves across an economy, it can influence productivity, market access, and the allocation of workers and firms.

This report asks whether countries with higher levels of internet access also have higher levels of economic development. The empirical focus is the relationship between internet penetration, measured as the percentage of the population using the internet, and GDP per capita. The question is important because digital infrastructure is often presented as a development priority. If internet access is strongly related to income, then investment in digital infrastructure may help countries participate more fully in global markets. At the same time, richer countries are also more able to build networks, buy devices, and support the education systems that make internet use productive. This creates a difficult empirical problem: the correlation between internet access and income may reflect both the effect of internet access on development and the effect of development on internet adoption.

The project therefore distinguishes descriptive association from causal interpretation. A simple scatter plot and pooled OLS regression capture the overall cross-country correlation. Additional specifications then introduce controls and fixed effects to compare countries more carefully over time. The controls account for observable differences in education, urbanization, and investment. Country fixed effects absorb time-invariant differences across countries, such as geography, long-run institutions, and historical development paths. Year fixed effects absorb global shocks and common time trends, including global technological change and macroeconomic cycles. Finally, an exploratory instrumental-variable specification uses lagged fixed telephone subscriptions as an instrument for internet use. This IV approach is useful for discussion, but it is not treated as definitive causal evidence because older telecommunications infrastructure may affect development through channels other than internet access.

The results show a clear pattern. Internet access is strongly and positively associated with GDP per capita in pooled regressions. The coefficient is much smaller after introducing controls and fixed effects, which suggests that much of the raw correlation reflects broader differences between countries rather than the independent effect of internet access. In the two-way fixed-effects specification, the internet coefficient remains positive but is weakly statistically significant. The IV estimate is larger, but its interpretation depends on a demanding exclusion restriction. The main conclusion is therefore cautious: internet access appears to move together with economic development, but this project cannot prove that internet expansion alone causes large increases in income.

2 Related Literature

This project relates to a large literature on telecommunications infrastructure, ICT adoption, and economic growth. One early contribution is Röller and Waverman [2001], who study telecommunications infrastructure and economic development using a simultaneous-equations approach. Their paper is important because it treats telecommunications investment and economic growth as jointly determined, rather than assuming that infrastructure is exogenous. This insight is directly relevant here: countries with higher GDP per capita may have better internet access because they can afford digital infrastructure, while digital infrastructure may also raise GDP per capita.

Later work focuses more directly on broadband and internet technologies. Czernich et al. [2011] analyze broadband infrastructure and economic growth in OECD countries, finding that broadband diffusion is positively associated with growth. Qiang et al. [2009] also emphasize broadband as a general-purpose technology that can reduce transaction costs, expand access to information, and support productivity. These mechanisms suggest that the internet may matter not only as a consumer good, but also as an input into production and market coordination.

At the microeconomic and sectoral level, the productivity effects of ICT are often linked to complementary investments. Cardona et al. [2013] review evidence showing that ICT can raise productivity, but that the gains often depend on organizational change, skills, and complementary intangible capital. This matters for the present study because internet access alone may not be sufficient. A country may have high internet penetration but limited gains if firms, workers, and institutions cannot use digital tools effectively.

The literature also highlights the value of credible identification strategies. Hjort and Poulsen [2019] study the arrival of fast internet in Africa and show how changes in access to submarine fiber-optic cables affected employment outcomes. Their setting is more credible for causal inference than a broad cross-country panel because the timing and geography of infrastructure arrival provide a clearer source of variation. By contrast, this report uses a global panel and fixed effects, which are useful for controlling for many confounders but still leave open concerns about reverse causality and omitted variables.

This project contributes by applying these ideas to a recent global World Bank panel covering 2005–2024. The goal is not to produce a definitive causal estimate, but to show how the estimated relationship changes as the empirical specification becomes more demanding. The analysis therefore emphasizes interpretation: large pooled correlations should not be read as causal effects, and even fixed-effects estimates require careful discussion of identification assumptions.

3 Data

3.1 Data Source and Sample Construction

The data come from the World Bank's World Development Indicators (WDI), the World Bank's main development-indicator database compiled from officially recognized international sources [World Bank, 2026a]. The cleaned dataset used in the main analysis covers the years 2005–2024 and includes 2,336 country–year observations from 168 countries. The panel is unbalanced because not every country reports every variable in every year. The analysis uses complete cases after excluding aggregate regions and dropping observations with missing values for the outcome, internet access, and the three main controls.

The dependent variable is the natural logarithm of GDP per capita, measured in PPP-adjusted constant 2021 international dollars. The main explanatory variable is internet access, defined in the WDI metadata as the percentage of individuals who used the internet from any location in the last three months. The controls are gross secondary school enrollment, urban population as a share of total population, and gross capital formation as a share of GDP. The exploratory IV sample additionally uses lagged fixed telephone subscriptions per 100 people as an instrument for internet access; WDI reports this series from International Telecommunication Union data [World Bank, 2026b]. Because the instrument is lagged, the IV sample begins in 2006 and contains fewer observations than the main sample.

Table 1: Variable Definitions

Variable	Definition	WDI Indicator
GDP per capita	PPP-adjusted GDP per capita, constant 2021 international dollars	`NY.GDP.PCAP.PP.KD`
Internet access	Individuals using the internet in the last three months (% of population)	`IT.NET.USER.ZS`
Education	Gross secondary school enrollment rate	`SE.SEC.ENRR`
Urbanization	Urban population (% of total population)	`SP.URB.TOTL.IN.ZS`
Investment	Gross capital formation (% of GDP)	`NE.GDI.TOTL.ZS`
Lagged telephone subscriptions	Fixed telephone subscriptions per 100 people, lagged	`IT.MLT.MAIN.P2`

Source: World Bank WDI metadata [World Bank, 2026b].

3.2 Descriptive Statistics

Table 2 presents summary statistics for the main estimation sample. The average GDP per capita is about $28,712, but the standard deviation is also large, showing that the sample combines low-, middle-, and high-income economies. The mean of log GDP per capita is 9.77. Internet penetration averages 52.65 percent of the population, with a standard deviation of 31.33 percentage points. This large spread is useful for estimation because the sample includes countries and years at very different stages of digital adoption.

Education, urbanization, and investment also vary substantially. The average gross secondary enrollment rate is 86.76 percent, but the maximum exceeds 100 percent because gross enrollment can include students outside the official age range. Urbanization averages 63.34 percent, indicating that the typical country–year observation in the estimation sample is majority urban. Investment averages 24.83 percent of GDP. These variables are included because they are plausibly related to both internet access and economic development: more educated, more urban, and more investment-intensive economies may both adopt internet technologies faster and have higher output per person.

Table 2: Summary Statistics

	Mean	SD	Min	Max	N
gdp_pc	28,711.80	26,973.34	952.12	174,569.52	2336
log_gdp_pc	9.77	1.11	6.86	12.07	2336
internet	52.65	31.33	0.10	100.00	2336
education	86.76	26.69	3.35	164.08	2336
urban	63.34	21.49	15.49	100.00	2336
investment	24.83	7.66	0.00	70.33	2336

3.3 Descriptive Relationship

Figure 1 plots internet access against log GDP per capita. The relationship is strongly positive: country–year observations with higher internet penetration tend to have higher income per person. This descriptive pattern is consistent with the idea that digital connectivity is part of economic development, but it does not establish causality. Richer countries may have more internet access because they have better infrastructure, higher education, stronger institutions, and greater household purchasing power. The rest of the analysis therefore asks whether the relationship survives more demanding controls.

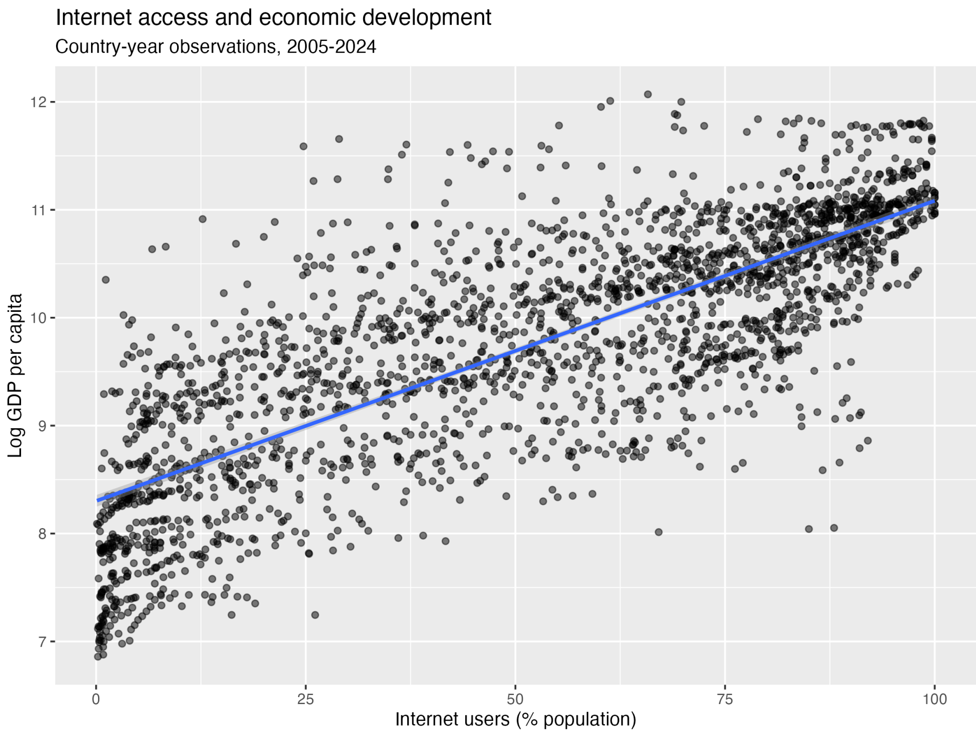

Figure 1: Internet Access and Log GDP per Capita, 2005–2024

4 Empirical Methodology

4.1 Baseline OLS

The first specification estimates the unconditional relationship between internet access and income:

$$\log(y_{it}) = \alpha + \beta Internet_{it} + \varepsilon_{it}, \tag{1}$$

where y_{it} is GDP per capita in country i and year t, and $Internet_{it}$ is the percentage of the population using the internet. Because the dependent variable is in logs and internet access is measured in percentage points, β is interpreted approximately as the percent difference in GDP per capita associated with a one percentage-point increase in internet access. For example, a coefficient of 0.001 implies that a 10 percentage-point increase in internet access is associated with about 1 percent higher GDP per capita.

Equation (1) is useful as a benchmark, but it is unlikely to be causal. It compares high- and low-internet observations without accounting for the many reasons why they differ. The coefficient may therefore combine the effect of internet access with the effects of education, urbanization, investment, institutions, and long-run development.

4.2 Controls

The second specification adds observable time-varying controls:

$$\log(y_{it}) = \alpha + \beta Internet_{it} + \gamma X_{it} + \varepsilon_{it}, \tag{2}$$

where X_{it} includes education, urbanization, and investment. Education captures human capital and the ability of workers and firms to use digital tools. Urbanization captures density, infrastructure, and market access. Investment captures capital accumulation and broader development dynamics. Adding these controls reduces omitted-variable bias if the controls are correlated with both internet adoption and GDP per capita.

However, controlled OLS still cannot eliminate all confounding. Countries differ in many persistent ways that are difficult to measure, including geography, legal systems, state capacity, colonial history, and long-run institutional quality. These factors may affect both digital infrastructure and income.

4.3 Fixed Effects

The third and fourth specifications introduce fixed effects. The country fixed-effects model is

$$\log(y_{it}) = \beta Internet_{it} + \gamma X_{it} + \alpha_i + \varepsilon_{it}, \tag{3}$$

where α_i is a country fixed effect. This specification compares each country with itself over time, rather than comparing different countries to each other. It removes time-invariant country characteristics that may otherwise bias the estimate.

The preferred specification adds year fixed effects:

$$\log(y_{it}) = \beta Internet_{it} + \gamma X_{it} + \alpha_i + \delta_t + \varepsilon_{it}, \tag{4}$$

where δ_t is a year fixed effect. Year fixed effects control for global shocks and common trends, such as the global financial crisis, the COVID-19 period, commodity price cycles, and worldwide technological change. Identification in equation (4) comes from within-country changes in internet penetration relative to global time patterns, conditional on the controls.

All models report standard errors clustered at the country level. Clustering is appropriate because repeated observations within the same country are likely to have correlated errors over time. Without clustering, the standard errors could be too small, making the results appear more precise than they really are.

4.4 Exploratory Instrumental-Variable Strategy

The main threat to causal interpretation is endogeneity. Internet access may increase GDP per capita, but higher GDP per capita may also increase internet access. In addition, unobserved reforms, infrastructure policies, or institutional improvements may affect both variables. To explore this issue, the project estimates a two-stage least squares model using lagged fixed telephone subscriptions as an instrument for internet access.

The first stage is

$$Internet_{it} = \pi_0 + \pi_1 Telephone_{i,t-1} + \lambda X_{it} + \alpha_i + \delta_t + \nu_{it}, \tag{5}$$

and the second stage is

$$\log(y_{it}) = \beta_{IV} \widehat{Internet}_{it} + \gamma X_{it} + \alpha_i + \delta_t + \varepsilon_{it}. \tag{6}$$

The relevance condition is plausible because countries with more fixed telephone infrastructure in the previous year are likely to have better communications networks and administrative capacity, making internet adoption easier. The exclusion restriction is more problematic. Lagged telephone penetration may be correlated with historical infrastructure investment, urbanization, state capacity, and business development, all of which can directly affect GDP per capita. For this reason, the IV model is presented as exploratory rather than as the preferred causal estimate.

5 Results

5.1 Main Regression Results

Table 3 reports the main estimates. Column 1 shows the baseline pooled OLS relationship. The coefficient on internet access is 0.028 and is statistically significant at the 0.1 percent level. Interpreted mechanically, this implies that a one percentage-point increase in internet access is associated with about 2.8 percent higher GDP per capita, so a 10 percentage-point increase is associated with about 28 percent higher GDP per capita. This estimate is too large to interpret causally because it mostly compares richer, highly connected economies with poorer, less connected economies.

Column 2 adds education, urbanization, and investment. The coefficient on internet access falls from 0.028 to 0.011, but remains highly statistically significant. This decline indicates that part of the raw internet–income relationship reflects other development characteristics. Education and urbanization are both positively associated with GDP per capita in this specification, suggesting that countries with more schooling and more urban populations also tend to be richer.

Column 3 adds country fixed effects. The internet coefficient falls further to 0.004. This is an important change because country fixed effects remove persistent cross-country differences. The result means that the estimated association is much smaller when the analysis compares a country to itself over time. The coefficient remains statistically significant, but the economic magnitude is more moderate: a 10 percentage-point increase in internet penetration is associated with roughly 4 percent higher GDP per capita.

Column 4 adds both country and year fixed effects. This is the most demanding main specification. The coefficient on internet access falls to 0.001 and is significant only at the 10 percent level. The estimate suggests that a 10 percentage-point increase in internet access is associated with roughly 1 percent higher GDP per capita, conditional on controls and fixed effects. The result is still positive, but it is much smaller and less precise than the pooled OLS estimate. This pattern is exactly what one would expect if much of the unconditional relationship reflects broader development differences across countries.

Table 3: Main Regression Results

	Baseline OLS	Controls	Country FE	Country + Year FE
Internet access	0.028***	0.011***	0.004***	0.001+
	(0.001)	(0.001)	(0.001)	(0.001)
Education		0.014***	0.003***	0.002*
		(0.002)	(0.001)	(0.001)
Urbanization		0.019***	0.007*	0.004
		(0.003)	(0.003)	(0.003)
Investment		0.006	0.004**	0.004*
		(0.004)	(0.001)	(0.001)
Observations	2336	2336	2326	2326
R^2	0.619	0.791	0.992	0.993
Adjusted R^2	0.619	0.791	0.991	0.992
Country FE	No	No	Yes	Yes
Year FE	No	No	No	Yes

+ p <0.1, * p <0.05, ** p <0.01, *** p <0.001
Standard errors clustered at the country level are reported in parentheses.

5.2 Interpretation of the Controls

The control variables provide additional context. Education is positive and statistically significant across the controlled specifications, although the coefficient becomes smaller after fixed effects are added. This is consistent with the idea that human capital is related to income and may also complement internet use. Investment is positive in the fixed-effects specifications, suggesting that capital formation is associated with higher GDP per capita within countries over time. Urbanization is positive in the pooled and country fixed-effects models, but becomes statistically insignificant after adding year fixed effects. This does not mean that urbanization is unimportant; rather, it means that the remaining within-country variation in urbanization, after controlling for other variables and common year shocks, is not precisely associated with income in this sample.

The very high R^2 values in the fixed-effects models should also be interpreted carefully. Country fixed effects explain a large share of the variation in GDP per capita because countries differ persistently in income levels. A high R^2 in this setting does not prove that the causal model is correct. It mainly shows that the model fits income levels well after allowing each country to have its own intercept.

5.3 Exploratory IV Results

Table 4 presents the exploratory IV estimate. The coefficient on instrumented internet access is 0.005 and is statistically significant at the 0.1 percent level. The first-stage F-statistic is 353.8, which indicates strong instrument relevance, meaning that lagged fixed telephone subscriptions are strongly correlated with internet penetration after controls and fixed effects. However, the first-stage statistic does not establish instrument validity. In terms of magnitude, the IV coefficient implies that a 10 percentage-point increase in internet penetration is associated with about 5 percent higher GDP per capita.

Although the IV estimate is larger than the two-way fixed-effects estimate, it should not be interpreted as definitive evidence of a causal effect. The instrument is likely relevant, but its exclusion restriction is debatable. Fixed telephone infrastructure may proxy for earlier economic modernization, government capacity, business networks, or urban infrastructure. These factors could directly affect GDP per capita even if internet access did not change. Therefore, the IV results are best read as a robustness and discussion exercise: they use an exploratory specification based on a historically related predictor of internet access, but they do not remove all endogeneity concerns.

6 Limitations

Several limitations affect the interpretation of the results. The first is reverse causality. Higher GDP per capita can increase internet access by raising household income, expanding firm demand for digital services, and giving governments more fiscal resources to invest in infrastructure. Fixed effects reduce some forms of bias, but they do not fully solve reverse causality if income growth and internet adoption occur together over time.

The second limitation is omitted-variable bias. The controls capture education, urbanization, and investment, but they do not capture all relevant policy and institutional changes. For example, regulatory reforms, competition in telecommunications markets, electricity reliability, political stability, and trade openness may affect both internet access and GDP per capita. If these factors change over time within countries and are correlated with internet adoption, the fixed-effects estimates may still be biased.

Table 4: Exploratory Instrumental-Variable Results

	Exploratory IV
Internet access (instrumented)	0.005***
	(0.001)
Education	0.003**
	(0.001)
Urbanization	0.000
	(0.003)
Investment	0.005**
	(0.001)
Observations	2165
R^2	0.993
Adjusted R^2	0.993
RMSE	0.096
Country FE	Yes
Year FE	Yes
IV	Lagged fixed telephone subscriptions
First-stage F-statistic	353.8

+ p <0.1, * p <0.05, ** p <0.01, *** p <0.001
Standard errors clustered at the country level are reported in parentheses.

The third limitation concerns measurement. Internet users as a percentage of the population measures access or use, but not connection quality, affordability, digital skills, or productive use. A country with widespread slow connections may not receive the same economic benefit as a country with high-speed broadband. Similarly, GDP per capita is a broad measure of average income and may not capture distributional outcomes, informality, or non-market welfare improvements.

The fourth limitation is sample selection from missing data. The final dataset uses complete cases, so countries and years with missing education, investment, or other variables are excluded. If missingness is related to development level or statistical capacity, the estimation sample may not represent all countries equally.

Finally, the IV strategy is imperfect. Lagged fixed telephone subscriptions satisfy the relevance condition strongly in the data, but relevance is not enough. The key requirement is that lagged telephone infrastructure affects current GDP per capita only through current internet access, after controls and fixed effects. This is difficult to defend because telephone infrastructure is itself part of a country's broader development history. A stronger causal design would require a more plausibly exogenous source of internet access, such as variation in submarine cable arrival, terrain-based rollout costs, or policy rules that affected internet infrastructure independently of growth trends.

7 Conclusion

This report examined the relationship between internet access and economic development using World Bank panel data for 2005–2024. The descriptive evidence shows a strong positive relationship between internet penetration and log GDP per capita. Countries and years with more internet users tend to be richer. However, the regression results show that this relationship becomes much smaller as the specification controls more carefully for confounding factors.

The preferred two-way fixed-effects model estimates an internet coefficient of 0.001, implying that a 10 percentage-point increase in internet access is associated with roughly 1 percent higher GDP per capita. This estimate is positive but weakly statistically significant. The exploratory IV model produces a larger coefficient of 0.005 and a strong first stage, but the instrument is not fully convincing because fixed telephone infrastructure may influence development through channels other than internet access.

The main lesson is that internet access is closely connected to economic development, but the size and causal interpretation of the relationship depend heavily on the empirical strategy. Simple cross-country comparisons overstate the likely independent effect of internet access because they combine digital adoption with many other dimensions of development. More credible specifications suggest a smaller, positive association.

From a policy perspective, the results support the view that digital connectivity can be part of a development strategy, especially when combined with education, investment, and complementary infrastructure. However, expanding internet access alone is unlikely to be sufficient. The economic payoff from digital technologies likely depends on skills, institutions, competition, electricity, affordability, and the ability of firms and households to use digital tools productively. Future research could improve on this project by using more detailed broadband quality measures, household or firm-level data, and stronger sources of exogenous variation in internet infrastructure.

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A Code and Replication Notes

The empirical outputs used in this report are generated from three uploaded R scripts. `code/01_data.r` downloads and cleans the WDI panel; `code/02_analysis.r` creates the summary table, scatter plot, and main regressions; and `code/03_iv_exploration.r` constructs the lagged telephone instrument and estimates the exploratory IV model.

The main cleaned dataset contains 2,336 observations from 168 countries over 2005–2024. The IV cleaned dataset contains 2,176 observations from 165 countries over 2006–2024 before fixed-effects estimation; the reported IV table uses 2,165 observations. Standard errors are clustered at the country level.